

From Endoscopic Image Sequences to a Simulable Soft-Tissue Digital Twin: Gaussian Surface Tracking and Force-Scale-Aware Mechanical Inversion

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Abstract

Endoscopic reconstruction estimates appearance and surface motion but does not by itself provide a mechanically valid, simulable tissue model. We propose a soft-tissue digital-twin framework that converts endoscopic images into a co-registered Gaussian surface, optical material representation, displacement field, and force-aware volumetric mechanics. A canonical anisotropic Gaussian field is reconstructed with depth-valid, instrument-excluded supervision and tracked by fixed primitive identities using graph regularization, constant-velocity initialization, and divergence rollback. An explicit Gaussian-to-tetrahedral interface transfers surface observations to a differentiable Neo-Hookean solver. Known forces permit modulus inversion; without force sensing, a derived force-scale relation limits inference to stiffness contrast. Each parameter is labelled measured, estimated, or assumed and accompanied by uncertainty. Evaluation combines two EndoNeRF sequences, two SCARED keyframes, 60 finite-element scenarios, six phantom acquisitions, and one clinical computed-tomography case. On a strict 28-frame holdout, tissue peak signal-to-noise ratio/structural similarity were 16.7 dB/0.564 for the present method and 36.6 dB/0.939 for EndoGaussian. Across 60 finite-element scenarios, median errors were 9.6% for background modulus, 32.0% for inclusion modulus, and 32.4% for contrast. Phantom contrast was **3.80×** versus **3.56×** by compression testing. The contribution is a traceable observation-to-simulation interface; absolute tissue modulus still requires measured force.

Keywords: digital twin, soft tissue, endoscopic video, elasticity inversion, parameter provenance, biomechanics

1 Introduction

Surgical data science, robot-assisted autonomy, and patient-specific pre-operative planning all require a *simulable* model of soft tissue: a scene that can be loaded into a physics simulator and driven forward in time to predict how the tissue will deform, contact, and respond to instrument

interaction. A visual reconstruction represents observed appearance and geometry, whereas a digital twin additionally requires a forward model that predicts deformation under new loads [1, 2]. This paper presents a pipeline that builds such a twin: it takes an endoscopic image sequence with auxiliary depth and instrument masks as input

and produces a physically simulable soft-tissue digital twin as output (Fig. 1).

Potential applications include simulation-based rehearsal of tissue manipulation, haptic training, and prediction of deformation outside the endoscopic field of view. These applications require not only geometric and appearance fidelity but also mechanically supportable parameter estimates with quantified uncertainty.

The core difficulty is that geometry and appearance are observable from images, while mechanical parameters (Young’s modulus E , Poisson ratio ν , viscosity) are not. Radiance measurements do not directly identify stress or elasticity. A mechanical parameter can only be obtained in two ways: (a) it is looked up from a tissue-class prior (an assumption), or (b) it is inferred by observing deformation under known loading (a measurement, subject to the identifiability limits of the loading). Prior reconstruction studies generally prescribe mechanical constants without quantifying their evidential basis. Predictions obtained with such constants cannot be interpreted as patient-specific mechanical estimates. Accordingly, the proposed twin records whether each parameter was measured, estimated, or assumed [1].

1.1 Related work

summarizes representative methods in medical digital twins, endoscopic reconstruction, and physics-based system identification.

1.1.1 Medical digital twins

Digital twins originated in industry, where a constantly updating virtual copy enables analysis, simulation, and prediction of a real-world asset [3, 4], and are increasingly studied as a route to precision medicine. Recent frameworks decompose a medical twin into the patient, the data connection, the patient-in-silico model, the interface, and twin synchronisation [1, 2]. The cardiology community has produced the most mature examples, with electrophysiology heart twins used to plan ablation and resynchronization therapy; oncology has produced tumor-growth twins coupling imaging with reaction-diffusion models; and epidemiology has produced population-level twins for policy testing. These share a common structure (a

mechanistic or learned patient-in-silico model synchronized with patient data) but their state variables are compartmental, hemodynamic, or cellular rather than mechanical. Intra-operative soft-tissue twins built from endoscopic video remain largely unexplored. Existing frameworks also do not specify the provenance of individual physical parameters. The present method incorporates a measured/estimated/assumed audit and uses the tracking loop (Sec. 2.9) for twin synchronisation [1].

1.1.2 Endoscopic scene reconstruction

EndoNeRF [5] introduced neural radiance fields [6] for deformable endoscopic scenes, demonstrating novel-view synthesis of tissue deformation by conditioning a radiance field on a learned deformation code per frame. EndoSurf [7] improved surface fidelity by combining a neural signed-distance field with a surface-aware rendering loss, yielding cleaner tissue boundaries. A line of Gaussian-splatting methods [8], including EndoGaussian [9], replaced the implicit field with anisotropic 3D Gaussians, gaining an order of magnitude in rendering speed and enabling per-primitive appearance. These methods reconstruct appearance and geometry but do not recover physical parameters, and their output cannot be loaded into a physics simulator. Complementary lines address camera tracking and mapping in endoscopy, from stereo correspondence [10] and robust pose estimation [11] to deformable SLAM [12], and inverse-rendering methods recover relightable appearance with bidirectional reflectance distribution function (BRDF) decomposition [13, 14], but do not estimate mechanical parameters. The present reconstruction uses the Gaussian representation with a constrained BRDF (Sec. 2.3); the same field is subsequently tracked and mechanically parameterized. Tool-aware masking restricts supervision to tissue pixels and improves the corresponding ablation by ~ 25 dB.

1.1.3 Physics-aware reconstruction

A growing line of work couples neural reconstruction with physics. PAC-NeRF [15] performs system identification from multi-view video with known forces, recovering material parameters of household objects by matching a differentiable continuum simulator to observed motion; its key

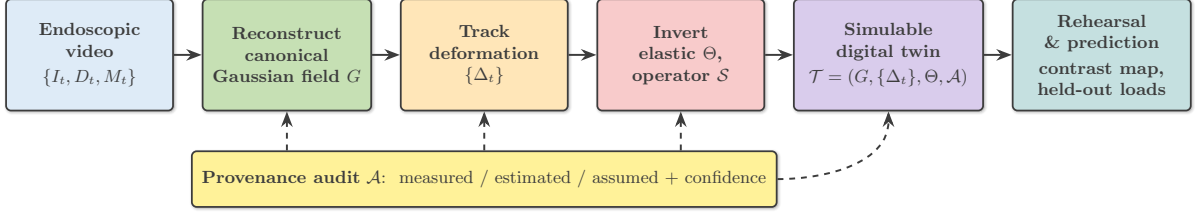


Fig. 1 Overall pipeline for converting an endoscopic image sequence and auxiliary depth into a soft-tissue digital twin that can be driven forward in a physics simulator. Reconstruction fits a canonical Gaussian field G from the reference frame; tracking estimates the deformation sequence $\{\Delta_t\}$ on that fixed set of primitives; and mechanical inversion recovers the elastic parameters Θ and the forward operator $\mathcal{S}$ by differentiable simulation. The provenance audit $\mathcal{A}$ accompanies the result, tagging every physical parameter as measured, estimated, or assumed and attaching an uncertainty estimate.

requirement is a measured force, which is rarely available in endoscopy. PhysDreamer [16] uses a video diffusion prior to estimate the relative stiffness of static objects, trading absolute accuracy for the ability to work from a single video. PhysGaussian [17] embeds the Material Point Method into Gaussians for generative dynamics but requires manually supplied material parameters, so the physics is imposed rather than recovered. Spring-Gaus and similar variants attach spring models to Gaussians for elastic dynamics [18, 19], and graph-network simulators learn complex physics from data [20]. These works demonstrate that physics can be coupled with learned reconstruction, but they either assume known forces, recover only relative stiffness, or require manual parameters. None of the compared methods reports per-parameter provenance. The present method targets endoscopic soft tissue, recovers absolute modulus when the applied load is known, and records the provenance of every parameter. the force-scale relation shows why the force scale constrains absolute-modulus recovery.

1.1.4 Elastography

Quasi-static elastography recovers modulus contrast from deformation under a shared-stress approximation [21–23], and is clinically established for liver fibrosis staging and tumor detection. The method tracks the axial strain induced by a small compression; under the assumption that the stress is approximately uniform across a region, the ratio of strains in two regions approximates the inverse ratio of their moduli, so a stiff inclusion appears as a low-strain region. Without a force measurement, this approximation identifies

contrast rather than absolute modulus. the force-scale relation provides the corresponding force-scale relation, and the audit therefore reports contrast when the force magnitude is unavailable.

1.1.5 Differentiable simulation and uncertainty

Differentiable physics (differentiable FEM, MPM, and mass-spring solvers) enables gradient-based parameter inversion through a simulator [15, 17, 20, 24]. We use a differentiable Neo-Hookean FEM for the main inversion and a fast mass-spring surrogate for synthetic closed-loop studies, and we quantify the surrogate’s inadequacy (82% forward error) to justify the FEM choice. Laplace approximations and Monte-Carlo propagation are established uncertainty methods [25–27]. Here they are used to identify weakly constrained parameters and to propagate parameter uncertainty to predicted displacement.

1.1.6 Methodological gap

The remaining gap is a co-registered framework that connects endoscopic reconstruction and temporally consistent surface displacement to mechanical parameterization while distinguishing measured quantities from prior assumptions. The proposed Gaussian field supplies the surface observations, differentiable Neo-Hookean inversion supplies the known-force mechanics, and the provenance map records their evidential status. Evaluation ranges from a synthetic closed loop to a physical phantom with independent compression measurements.

Based on 3D Gaussian splatting [8] and differentiable Neo-Hookean finite elements [24, 28], we

propose an audited soft-tissue digital twin $\mathcal{T} = (G, \{\Delta_t\}, \Theta, \mathcal{A})$ (Fig. 1).

The representation couples appearance and motion through a canonical field of anisotropic Gaussians. Each primitive stores geometry and a constrained Cook–Torrance reflectance model [29]. Instrument-excluded, depth-valid supervision fits this field, and fixed primitive identities provide correspondences for estimating displacement over time. Constant-velocity initialization, graph regularization, and rollback after divergence stabilize the tracked displacement.

Mechanical parameterization is conditioned on the available force evidence. For uninstrumented endoscopic sequences, tracked Gaussians are partitioned into spatial regions and their mean displacement magnitudes provide a common-load relative-stiffness estimate; no absolute modulus is assigned from video. For benchmark cases with known forces, surface displacements are compared with a differentiable tetrahedral FEM solution, and Young’s modulus is recovered by implicit differentiation of quasi-static equilibrium. The Gaussian field and tetrahedral mesh are separate discretizations coupled through surface-displacement observations. Thus, the shared representation refers to co-registration of appearance, motion, and regional mechanical metadata rather than use of Gaussian primitives as FEM elements.

The force-scale analysis formalizes the corresponding identifiability limit. At fixed Poisson ratio, scaling all regional moduli and the applied force by the same factor leaves equilibrium displacement unchanged (the force-scale relation); absolute modulus therefore inherits force-scale uncertainty, whereas regional stiffness contrast does not. An auxiliary in-house R3D-18 retraction-force estimator supplies an empirical force-error model for sensitivity analysis. It is external to the digital-twin framework and is not claimed as a contribution.

A log-normal Bayesian update records each parameter as measured, estimated, or assumed and propagates posterior uncertainty through the simulator (Sec. 2.8). Direct sensor observations are measured, reconstruction or inverse-simulation outputs are estimated, and table-derived values without case-specific evidence are assumed.

The principal contribution is a continuous image-to-simulation chain rather than a new rendering backbone. Endoscopic images are converted

into a canonical Gaussian surface with decomposed optical material, fixed primitive identities preserve surface correspondence during deformation, and an explicit surface-to-volume map supplies observations to the tetrahedral mechanics model. This co-registration enables rendered appearance, regional material metadata, tracked displacement, and simulated stress and strain to refer to the same tissue regions. The force-scale analysis then determines whether the available evidence supports absolute modulus or only relative contrast, while the provenance map preserves that distinction in the exported twin. The renderer, constitutive law, and Bayesian update are established techniques; the new contribution is their mechanically explicit, identifiability-aware coupling from endoscopic observation to forward simulation.

The evaluation uses two EndoNeRF sequences, two SCARED keyframes, all 60 scenarios and four load cases of the FEM benchmark, six ultrasound-phantom acquisitions with compression-test references, and one patient-specific liver geometry derived from 82 de-identified abdominal computed tomography (CT) slices. The Gaussian surface is also mapped directly to a tetrahedral volume to test the computational interface between optical and mechanical discretizations.

On the common 28-frame holdout, the present representation reaches 16.7 dB tissue-masked PSNR and 0.564 structural similarity, whereas the specialized EndoGaussian novel-view-synthesis method reaches 36.6 dB and 0.939. This gap is reported explicitly: the present framework preserves primitive identity for mechanical correspondence and does not claim state-of-the-art reconstruction. Tracking remains stable where the inertia-only ablation diverges. Across all 60 FEM scenarios, median relative errors are 9.6% for background modulus, 32.0% for inclusion modulus, and 32.4% for stiffness contrast. The ultrasound phantom gives 6.9% contrast error against compression testing. Absolute modulus is not reported for real tissue without measured force.

The rest of the paper follows the journal’s structure: Section 3 reports the results, Section 4 discusses implications and limitations, Section 2 details the methods, and Section 5 concludes.

2 Methods

The study was designed as a computational methods investigation of whether an endoscopic reconstruction with auxiliary depth can be extended to a mechanically simulable representation with explicit parameter provenance. Evaluation combines public endoscopic and phantom data, a synthetic FEM benchmark, and one de-identified clinical CT case. The following subsections define the representation, reconstruction, deformation tracking, mechanical inversion, force-scale analysis, and provenance map.

2.1 Study design and data sources

Datasets.

Table 1 summarizes the five data sources and their roles.

EndoNeRF frames are loaded at $0.5\times$ scale, with depth used both for the canonical point cloud and for the tissue mask Ω of (4) [5]; SCARED keyframes at $0.5\times$ scale with intrinsics from the calibration file [10]; the FEM benchmark is used with the measured (not true) forces and the noisy (not true) surface motion as inversion inputs; all 60 FEM scenario directories are inverted in deterministic lexical order with a frozen 20-iteration protocol; and the ÉTS phantom is used at the frame level with synchronized force [30].

Evaluation metrics.

Reconstruction quality is reported as PSNR and structural similarity (SSIM) on a strict common holdout, in which frames satisfying $(t - 1) \bmod 8 = 0$ are excluded from training for both methods. Full-image and tissue-masked metrics use the same source images and masks. Tissue PSNR is computed from squared error on selected pixels; tissue SSIM is the mean of the 11-by-11 local SSIM map over selected tissue pixels, so unselected black regions do not inflate the score. Canonical fit PSNR is reported separately and is not interpreted as held-out performance. Tracking quality is measured by the per-frame and worst-frame photometric loss. Mechanical accuracy is measured by (i) the relative error of the recovered modulus, $|E^{est} - E^{gt}|/E^{gt}$; (ii) the inclusion-contrast ratio E_{incl}/E_{bg} against ground truth; (iii) the held-out load-case displacement error $\|u(E^{est}) - u^{obs}\|/\text{mean}\|u^{obs}\|$; and (iv) the

95% prediction-interval coverage of held-out surface nodes.

Comparison methods.

For reconstruction, the primary comparison reruns the official EndoGaussian implementation [9] and the present method under the same one-in-eight holdout and evaluator. Published EndoNeRF [5] and EndoSurf [7] values are used only as context when their protocols differ. For mechanics we compare against the system-identification setting of PAC-NeRF [15], the relative-stiffness setting of PhysDreamer [16], and controlled ablations (per-node field versus two-region, mass-spring surrogate versus FEM-in-the-loop).

Implementation.

Reconstruction uses 800–1000 Adam iterations and approximately 6.5–7.7 thousand Gaussians on an RTX 5090 with the gsplat backend (~ 21 s per scene). Tracking uses 40–50 iterations per frame. The full FEM campaign uses 20 inversion iterations per scenario and requires 18.7 min per scenario on average; scenarios are independent and were distributed across available workers.

Statistical analysis.

Results are summarized with per-scene values, means, medians, and relative errors as specified for each experiment. The phantom median is accompanied by a nonparametric bootstrap 95% confidence interval. No hypothesis tests or power calculations were performed because the experiments evaluate algorithmic accuracy on fixed benchmark cases rather than estimate a clinical population effect. All reported values are generated by the released scripts (Sec. 4).

2.2 Soft-tissue digital twin

Definition 1 (Soft-tissue digital twin). A *soft-tissue digital twin* is a tuple $\mathcal{T} = (G, \{\Delta_t\}, \Theta, \mathcal{A})$, where G is a canonical geometric and optical scene representation, $\{\Delta_t\}$ is a deformation sequence on G , Θ is a set of physical parameters (elastic, viscous, inertial, optical), and $\mathcal{A}$ is an audit map assigning to each $\theta \in \Theta$ a provenance tag and confidence. The twin is simulable if there exists a forward operator $\mathcal{S} : (G, \Theta, f) \mapsto u$ mapping a load f to a deformation u , and it is audited if $\mathcal{A}$ is complete (every θ is tagged).

Table 1 Data sources. “GT” indicates the ground truth used for evaluation.

Dataset	Modality	Scale	GT used	Used for
EndoNeRF (2 scenes)	RGB sequence, auxiliary depth, instrument masks	63 / 156 frames	depth, instrument masks	reconstruction and tracking
SCARED	stereo laparoscopy + metric depth	2 keyframes	metric depth (mm)	reconstruction (portability)
FEM bench-mark	synthetic FEM (Neo-Hookean tetrahedra)	60 scenarios, 4 loads	E_{bg} , E_{incl} , forces	mechanical inversion (closed loop)
ÉTS phantom	ultrasound + force	6 acquisitions	compression-test E	mechanical inversion (real phantom)
Hospital CT	abdominal CT (1 patient)	82 slices	CT anatomy	geometry construction

The input is a sequence of endoscopic frames $\{I_t\}_{t=1}^T$ with per-frame depth maps D_t and tool masks M_t ; the output is the twin $\mathcal{T}$. As shown in , the canonical Gaussian field G is reconstructed, displaced while preserving primitive identity, and coupled to a mechanical forward operator. The audit map is attached to the resulting parameters. illustrates the result on an endoscopic sequence, and defines the notation.

2.3 Canonical scene reconstruction

Reconstruction builds G . Each Gaussian g_i stores a position $\mathbf{x}_i \in \mathbb{R}^3$, an anisotropic covariance $\Sigma_i = Q(q_i) \text{diag}(s_i^2) Q(q_i)^\top$ with rotation quaternion q_i and scale s_i , an opacity $o_i \in [0, 1]$, a unit normal $\mathbf{n}_i$, and BRDF parameters $(\rho_d, \rho_s, \alpha, m, a)$. A soft-max budget regularizes the relative lobe weights by a partition of unity,

$$\rho_d + \rho_s + a = 1, \quad \rho_d, \rho_s, a \geq 0, \quad (1)$$

which prevents unconstrained growth of these fitted coefficients; it is not by itself a proof of global energy conservation under the discretized lighting model. Images are formed by front-to-back alpha compositing,

$$\hat{I} = \sum_{i=1}^N c_i \alpha_i \prod_{j<i} (1 - \alpha_j), \quad (2)$$

where c_i is the shaded color of g_i and α_i its projected opacity. The shaded color follows a Cook–Torrance microfacet BRDF [29] split into a

diffuse and a specular lobe,

$$c_i = \underbrace{\frac{\rho_d}{\pi} E_d}_{\text{diffuse}} + \underbrace{\rho_s \frac{D(\alpha) G_s F}{4 (\mathbf{n} \cdot \mathbf{l})(\mathbf{n} \cdot \mathbf{v})}}_{\text{specular}}, \quad (3)$$

where E_d is the diffuse irradiance, $D(\alpha)$ the GGX normal distribution, G_s the Smith geometry term, F the Schlick Fresnel term, $\mathbf{l}$ the light direction, and $\mathbf{v}$ the view direction. The dot products in the denominator are clamped away from zero on visible front-facing samples, and the specular term is zero for back-facing samples. The highlight-free diffuse image is obtained by setting the specular lobe to zero.

A key endoscopic adaptation concerns the supervision mask. The EndoNeRF **masks**/ channel marks *tools*, not tissue, and tools have no valid depth. We therefore define the tissue pixel set as

$$\Omega = \{(u, v) : D(u, v) > \epsilon\} \setminus M, \quad (4)$$

with ϵ a small depth-validity threshold, and supervise only on Ω . Supervising on the tool mask instead of Ω degrades reconstruction by ~ 25 dB (Table 6). The reconstruction objective is

$$\mathcal{L}_1 = \lambda_{ph} \mathcal{L}_{photo} + \lambda_d \mathcal{L}_{depth} + \lambda_\alpha \mathcal{L}_\alpha, \quad (5)$$

with non-negative weights $\lambda_{ph}, \lambda_d, \lambda_\alpha$, a masked $L_1 + 0.5 L_2$ photometric term $\mathcal{L}_{photo}$ on Ω , a depth L_1 term $\mathcal{L}_{depth}$ on Ω , and an alpha binary-cross-entropy term $\mathcal{L}_\alpha$ toward the tissue mask. Lighting is an endoscope-co-located directional light plus a strong ambient spherical-harmonic lobe, so appearance approximates albedo; a learnable exposure closes the residual gap.

2.4 Deformation tracking

Tracking estimates $\{\Delta_t\}$ as a per-Gaussian displacement $\Delta_t \in \mathbb{R}^{N \times 3}$ on the canonical topology, keeping correspondences fixed across time. Fixed correspondence is what distinguishes a twin from a per-frame rebuild: the same primitive is followed through time, so its trajectory is a boundary condition. Naive inertia-only tracking (penalizing $\|\Delta_t - \Delta_{t-1}\|^2$) occasionally diverges on ambiguous frames. We therefore optimize

$$\Delta_t^* = \arg \min_{\Delta} \mathcal{L}_{photo}(\Delta) + \lambda_s \sum_{i=1}^N \|\Delta_i - \bar{\Delta}_{\mathcal{N}(i)}\|^2 + \lambda_a \|\Delta - 2\Delta_{t-1} + \Delta_{t-2}\|^2, \quad (6)$$

initialized from the constant-velocity extrapolation $\Delta_t^{(0)} = 2\Delta_{t-1} - \Delta_{t-2}$. The second term is a k -nearest-neighbor Laplacian smoothness over the Gaussian graph; the quantity $\bar{\Delta}_{\mathcal{N}(i)}$ is the mean displacement of the neighbors of primitive i . The third term is an acceleration prior that penalizes changes in velocity rather than velocity itself. A per-frame step cap constrains $N^{-1} \sum_i \|\Delta_{t,i}\|_2$ to at most three times the running median of preceding displacement magnitudes, and a divergence rollback repeats optimization at half the learning rate when the final loss exceeds three times the preceding-frame loss. If the criterion remains unmet, Δ_{t-1} is retained and the frame is flagged. The output is a sequence of replayable boundary conditions.

2.5 Coupling tracked Gaussians to mechanics

The optical and mechanical computations use different discretizations. Gaussian primitives represent the observed surface, whereas the FEM benchmark uses a tetrahedral volume mesh. Coupling is therefore performed through surface displacement rather than by treating Gaussians as finite elements.

For an uninstrumented endoscopic sequence, canonical Gaussian positions are partitioned into four spatial regions C_r by k -means. For region r , the mean displacement magnitude is

$$d_r = \frac{1}{|C_r|} \sum_{i \in C_r} \|\Delta_{t,i}\|_2. \quad (7)$$

Under the common-load approximation used in quasi-static elastography, a relative-stiffness proxy is

$$\kappa_r = \frac{\max_s d_s}{d_r + \epsilon_d}, \quad (8)$$

where $\epsilon_d = 10^{-9}$ in the displacement units prevents division by zero and the most compliant region has $\kappa = 1$. This calculation provides regional contrast only. The absolute modulus remains the tissue prior because the endoscopic sequence does not contain a force measurement.

For known-force benchmark experiments, the observed displacement is defined on surface nodes Γ_{obs} of the tetrahedral mesh. The FEM parameters are optimized so that simulated surface displacement matches these observations. The two settings evaluate complementary outputs: co-registered contrast on tracked Gaussians and absolute-modulus inversion on a force-calibrated volumetric mesh.

For the patient-specific geometry experiment, a binary CT liver mask is block-reduced and each occupied voxel is divided into six tetrahedra with shared nodes. The Gaussian surface is initialized to the volume boundary by principal-axis alignment and refined by trimmed similarity iterative closest point registration. For every covered volume-boundary node, the four nearest transformed Gaussians define inverse-distance weights w_{ij} , and tracked displacement is transferred as

$$u_j^{obs} = \sum_{i \in \mathcal{N}_4(j)} w_{ij} \Delta_i, \quad (9)$$

$$w_{ij} = \frac{(d_{ij} + \epsilon)^{-1}}{\sum_{k \in \mathcal{N}_4(j)} (d_{kj} + \epsilon)^{-1}}.$$

Only boundary nodes whose nearest Gaussian lies within 5% of the volume diagonal are marked as observed. This map preserves separate optical and mechanical discretizations while providing explicit correspondences for boundary conditions. Because the Gaussian and CT surfaces are from different subjects and the CT spacing metadata are unavailable, the experiment validates the computational interface, not anatomical registration.

2.6 Mechanical parameter inversion with known force

For known-force cases, mechanical inversion defines Θ and the forward operator $\mathcal{S}$. Soft tissue is modeled as a quasi-static continuum in equilibrium,

$$R(u; E, \nu, f) = f_{int}(u; E, \nu) - f = 0, \quad (10)$$

where f_{int} is the internal elastic force and f is the applied load. We invert E by gradient descent through a differentiable forward solver,

$$E^* = \arg \min_E \sum_t \|u(E, \nu, f_t) - u_t^{obs}\|_{\Gamma_{obs}}^2 + \gamma \mathcal{R}(E), \quad (11)$$

where $u(E, \nu, f_t)$ is the simulated displacement under load f_t , u_t^{obs} the observed deformation, $\gamma \geq 0$ a regularization weight, and $\mathcal{R}(E)$ a regularizer (log-normal prior plus spatial smoothness). Two forward models are used: a mass-spring lattice (structural and shear springs with stiffness $k = Eh$, h the lattice spacing) for synthetic closed-loop studies, and a differentiable Neo-Hookean FEM [28, 31, 32] with strain-energy density

$$\Psi(F) = \frac{\mu}{2} (\text{tr}(F^\top F) - 3) - \mu \log J + \frac{\lambda}{2} (\log J)^2, \quad (12)$$

where F is the deformation gradient, $J = \det F$, and $\mu = E/(2(1 + \nu))$, $\lambda = E\nu/((1 + \nu)(1 - 2\nu))$ are the Lamé parameters. The FEM solve uses Newton iteration with load stepping and warm starts, and gradients are obtained by the implicit-function theorem through the equilibrium. A kPa-scale log-normal tissue database (liver, fat, muscle, skin, tumor;) with near-incompressible ν and constitutive-model tags (Neo-Hookean / Mooney-Rivlin [33] / Ogden [31]) provides priors; the tissue class is selected by heuristic optical features (colour and specular response). This classification selects a prior category and does not directly determine the mechanical values.

2.7 Force-scale propagation

The following result provides the theoretical basis for reporting stiffness contrast as a force-scale-independent estimated quantity.

At fixed Poisson ratio, the Neo-Hookean internal force is linear in Young’s modulus. Scaling

all regional moduli and the applied force by the same factor therefore leaves equilibrium displacement unchanged, while regional stiffness contrast is invariant to that scale.

2.8 Parameter provenance

Definition 2 (Parameter provenance). *A physical parameter θ carries a provenance tag $\tau(\theta) \in \{\text{measured, estimated, assumed}\}$ and a confidence $c(\theta) \in [0, 1]$. A parameter is measured when it is acquired directly by a calibrated sensor, estimated when it is inferred by reconstruction or inverse simulation, and assumed when it is taken from a tissue-class table without case-specific evidence.*

Young’s modulus uses a log-normal prior $\log E \sim \mathcal{N}(\mu_0, \sigma_0^2)$, appropriate because tissue stiffness spans orders of magnitude and must stay positive. A Gaussian–Gaussian update combines the prior with deformation evidence in closed form,

$$\begin{aligned} \sigma_{post}^{-2} &= \sigma_0^{-2} + \sigma_{obs}^{-2}, \\ \mu_{post} &= \sigma_{post}^2 \left(\frac{\mu_0}{\sigma_0^2} + \frac{\mu_{obs}}{\sigma_{obs}^2} \right). \end{aligned} \quad (13)$$

where $(\mu_{obs}, \sigma_{obs})$ summarize the deformation evidence. The confidence is the fractional posterior narrowing, $c = 1 - \sigma_{post}/\sigma_0$. For uncertainty propagation, a Laplace approximation $p(\log E \mid \text{data}) \approx \mathcal{N}(\log E^*, H^{-1})$ with H the Hessian of the negative log-posterior at the optimum is sampled by Monte Carlo through the simulator to a 95% prediction interval.

2.9 Twin synchronization and simulation

A digital twin is not a one-shot reconstruction but a continuously updated copy [1]. Synchronization is the tracking loop: as each new frame arrives, the tracker updates $\{\Delta_t\}$ on the fixed canonical topology, and mechanical inversion can reuse the accumulated deformation as additional inversion evidence. Once constructed, the twin is used by driving $\mathcal{S}$ with new loads: given a planned instrument trajectory $\{f_t^{plan}\}$, the simulator produces the predicted deformation $\{u(E^*, \nu, f_t^{plan})\}$, which can be rendered through the canonical field for visual rehearsal or post-processed for quantities of interest (contact force, strain energy

density, margin strain). The audit modulates how these predictions are presented: predictions are accompanied by intervals propagated from the corresponding parameter uncertainties.

Remark 1 (Optimization and learned components). *The Gaussian field is optimized as a set of per-primitive parameters, the renderer is differentiable, and mechanical inversion optimizes physical parameters through the simulator. The auxiliary R3D-18 force estimator is external to the twin and is used only to define the force-error model in Sec. 3.3.*

2.10 Use of generative AI tools

During manuscript preparation, the authors used Cursor’s AI-assisted editor for English-language editing and LaTeX organization. Generative AI was not used to generate data, conduct experiments, calculate metrics, or draw scientific conclusions. The authors reviewed and approved all text and remain responsible for the manuscript.

3 Results

Reconstruction and tracking are reported on endoscopic and stereo video. Mechanical inversion is then reported as the force evidence changes, from a known load, to a video-based force estimate, to no force measurement. Uncertainty, comparison with related methods, and ablations follow. Limitations are discussed with the clinical implications.

3.1 Canonical reconstruction

The fitted Gaussian field is the representation that tracking and inversion reuse, so its tissue fidelity is reported first.

EndoGaussian exceeds the present method by 19.88 dB tissue PSNR and 0.375 tissue SSIM. EndoGaussian uses approximately 30,000 Gaussians and a learned spatiotemporal deformation field, whereas the present method uses approximately 6,500–6,800 fixed-identity primitives and interpolates tracked displacement at held-out times. The latter design maintains mechanical correspondence but is not competitive with a specialized dynamic novel-view-synthesis model. All-image PSNR is additionally affected by tools that the present representation deliberately excludes.

Table 3 reports per-scene reconstruction quality, and Fig. 3 shows the corresponding reconstructions. The higher PSNR in the pulling scene coincides with greater tissue coverage and more uniform illumination. In both EndoNeRF scenes, the largest errors occur near the instrument–tissue boundary. Each SCARED keyframe is fitted from one view without multi-frame supervision; these experiments assess transfer to a different anatomy, camera, and metric scale (Fig. 3).

3.2 Deformation tracking

Tracking attaches a displacement to each primitive of the field fitted above. Fixed primitive identities are what allow the sequence to be used as a boundary condition.

Tracked displacement concentrates at the tool contact region. Table 4 ablates the temporal model: the inertia-only baseline fails on frame 5 of the pulling scene. This failure coincides with a fast, partially occluded retraction and a weak local photometric gradient. Constant-velocity initialization and acceleration regularization avoid divergence, while rollback prevents the failed update from propagating to the next frame. On the cutting scene the same tracker runs six frames without divergence (loss 0.52–0.71). The tracked displacements are the observations passed to the mechanical inversion.

3.3 Mechanical inversion

Table 5 summarizes inversion as the force evidence changes: a known load, a video-based force estimate, and no force measurement. For the EndoNeRF sequences, tracked Gaussians produce a four-region relative-stiffness field through Eq. (8). Because these sequences provide neither force measurements nor stiffness references, this output is not assigned an absolute scale and is not used as a quantitative validation result. Absolute-modulus validation is performed on the force-calibrated FEM benchmark, and contrast validation is performed on the compression-tested phantom.

Synthetic closed loop

Uniform modulus is recovered to 0.01% error; a 4× stiff inclusion is recovered at 3.19× contrast from surface-only observations with two load cases.

Table 2 Strict one-in-eight EndoNeRF holdout. Both methods use the same 28 held-out frames, tissue masks, and evaluator.

Method	Frames	Full PSNR	Full SSIM	Tissue PSNR	Tissue SSIM
Present method	28	11.71	0.436	16.71	0.564
EndoGaussian	28	16.61	0.798	36.59	0.939

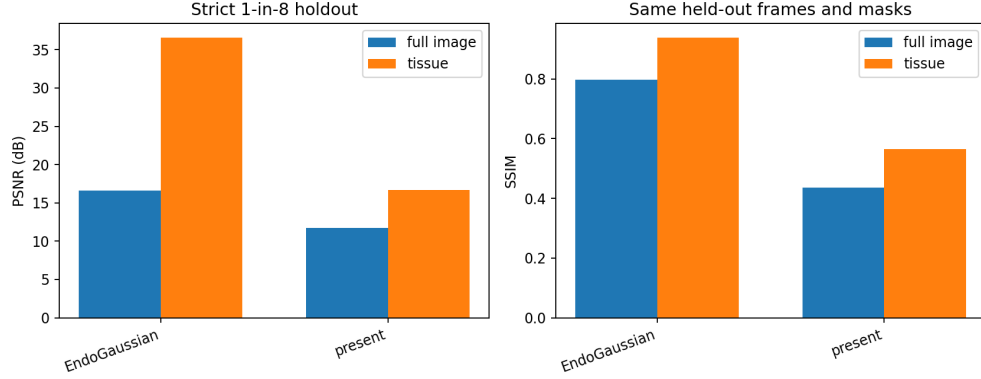


Fig. 2 Common-protocol reconstruction comparison. The present digital-twin pipeline prioritizes fixed surface correspondence and mechanical coupling; it does not match the specialized dynamic reconstruction baseline.

Independent FEM benchmark

With measured forces (5% calibration error) and noisy surface motion, all 60 predefined scenarios were inverted using the same frozen 20-iteration protocol. Background-modulus relative error had a median of 9.6%, a mean of 10.0%, and an interquartile range of 4.9–14.1%. Inclusion-modulus error was substantially larger (median 32.0%, mean 30.1%), as was stiffness-contrast error (median 32.4%, mean 33.8%). The complete campaign therefore supports moderate recovery of background stiffness but does not support accurate recovery of a deep inclusion or its contrast. Material inverted on `press_left` predicts unseen load cases with displacement error 11.1% (`press_right`), 3.7% (`shear_x`), and 14.9% (`shear_y`), against oracle-material errors of 9.6%, 7.2%, and 10.9%, respectively.

Force provenance

We use the empirical error model of an auxiliary in-house video-based force estimator (R3D-18, small-bowel retraction recordings, 6373 samples / 50 recordings: mean absolute error (MAE) 0.295 N, systematic scale factor 0.949 ± 0.140) and transport it onto the FEM benchmark. A systematic sweep (force scale 0.80–1.20, recovered E_{bg} ratio 0.846–1.110) follows the force-scale relation

in direction and is not an exact one-to-one map, as the numerical sensitivity analysis states. With this error model, the E_{bg} error is $4.1 \pm 4.0\%$ and the stiffness contrast stays within 2.22–2.27 $\times$.

Real phantom

On the ÉTS indentation phantom (6 acquisitions), the image-derived modulus contrast reaches a median of $3.80\times$ (bootstrap 95% confidence interval (CI) [3.16, 4.34]) against compression-test truth $3.56\times$ (background 31.0 ± 1.87 kPa, inclusion 8.71 ± 3.80 kPa), a 6.9% relative error, stable across 9 sensitivity configurations (3.5–9.3%). Posterior intervals contain both ground-truth moduli. The estimate is a contrast, not an absolute modulus, and the force-scale relation explains why this ratio is independent of a global force scale. Fig. 5 shows the per-acquisition estimates.

Hospital CT geometry construction

To assess construction of a patient-specific geometry from clinical data, we reconstruct the liver from an abdominal CT scan of a patient from Henan Provincial People’s Hospital. A right-upper-quadrant liver region is segmented on 82 axial slices and meshed into a patient-specific liver geometry (178,000 vertices). The CT provides a

Table 3 Canonical fit quality (tissue-masked PSNR); these values measure representation fit and are not held-out novel-view results.

Scene	Frames used	Tissue coverage	Gaussians	Tissue PSNR (dB)
EndoNeRF pulling	16–24	70.7%	6498	36.79
EndoNeRF cutting	16–24	74.1%	6824	34.09
SCARED keyframe_1	1	78.4%	7221	26.99
SCARED keyframe_2	1	83.1%	7676	26.17

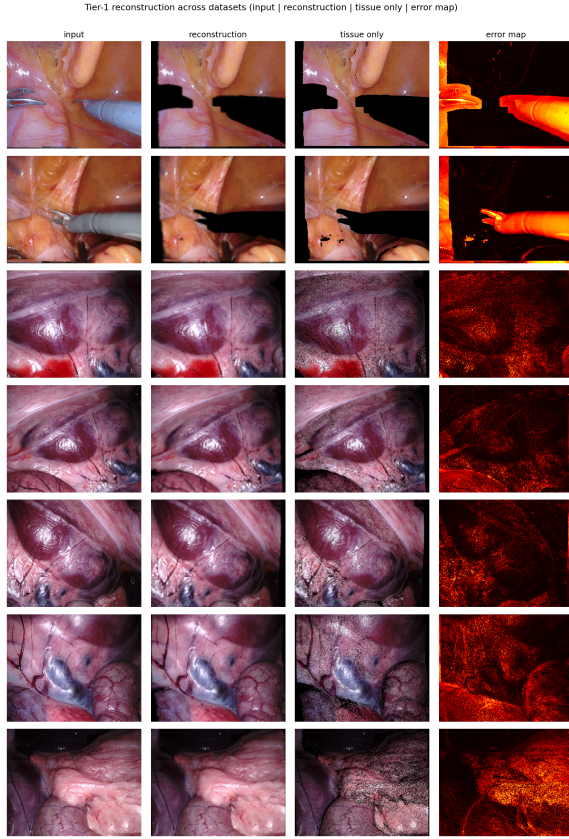


Fig. 3 Qualitative reconstruction. Rows are EndoNeRF pulling, EndoNeRF cutting, and SCARED keyframes. Columns are input, reconstruction, tissue only, and an error map. Tissue-masked PSNR is reported in Table 3 for the two EndoNeRF scenes and SCARED keyframes 1 and 2. The remaining SCARED rows are shown only as a visual check; they are not given a separate PSNR in this paper.

Table 4 Temporal-model ablation (pulling scene).

Tracker	Worst-frame loss	Behaviour
Inertia-only	37.23	frame 5 divergence
Constant velocity with roll-back	1.55	frame 5 rolled back

patient-specific geometry source; the absolute volume is not reported because the available slices do not contain spacing metadata. This CT case is not paired with an endoscopic sequence and therefore does not validate optical classification or mechanical inversion.

3.4 Posterior uncertainty

Posterior dispersion was examined as an indicator of parameter identifiability. The Laplace approximation gives $\text{std}(\log E_{bg}) = 1.17$ but a much larger $\text{std}(\log E_{incl}) = 21.2$: the data do not constrain the deep inclusion modulus, consistent with its 30% estimation error. The 95% prediction interval covers 100% of held-out surface nodes but is dominated by the unidentified inclusion direction, so the inclusion modulus is classified as weakly identified.

3.5 Ablations

The same Gaussians are used for rendering and deformation tracking on the endoscopic sequences; their displacements support regional contrast estimation. On the separate force-calibrated FEM benchmark, background modulus has a median error of 9.6% across 60 scenarios when the force is measured. A separate sensitivity analysis applies the auxiliary video-estimator error model; it is not a validation of the core framework. When no force is available, the reported quantity is stiffness contrast (6.9% error, interval covering the compression-test value), in agreement with the force-scale relation. The Laplace posterior marks the inclusion modulus as unidentified, and parameters copied from the tissue table are tagged assumed.

4 Discussion

Geometry and reflectance are constrained directly by image and depth observations. In contrast, Young’s modulus is identifiable only when both

FEM inversion campaign: 60/60 scenarios (complete)

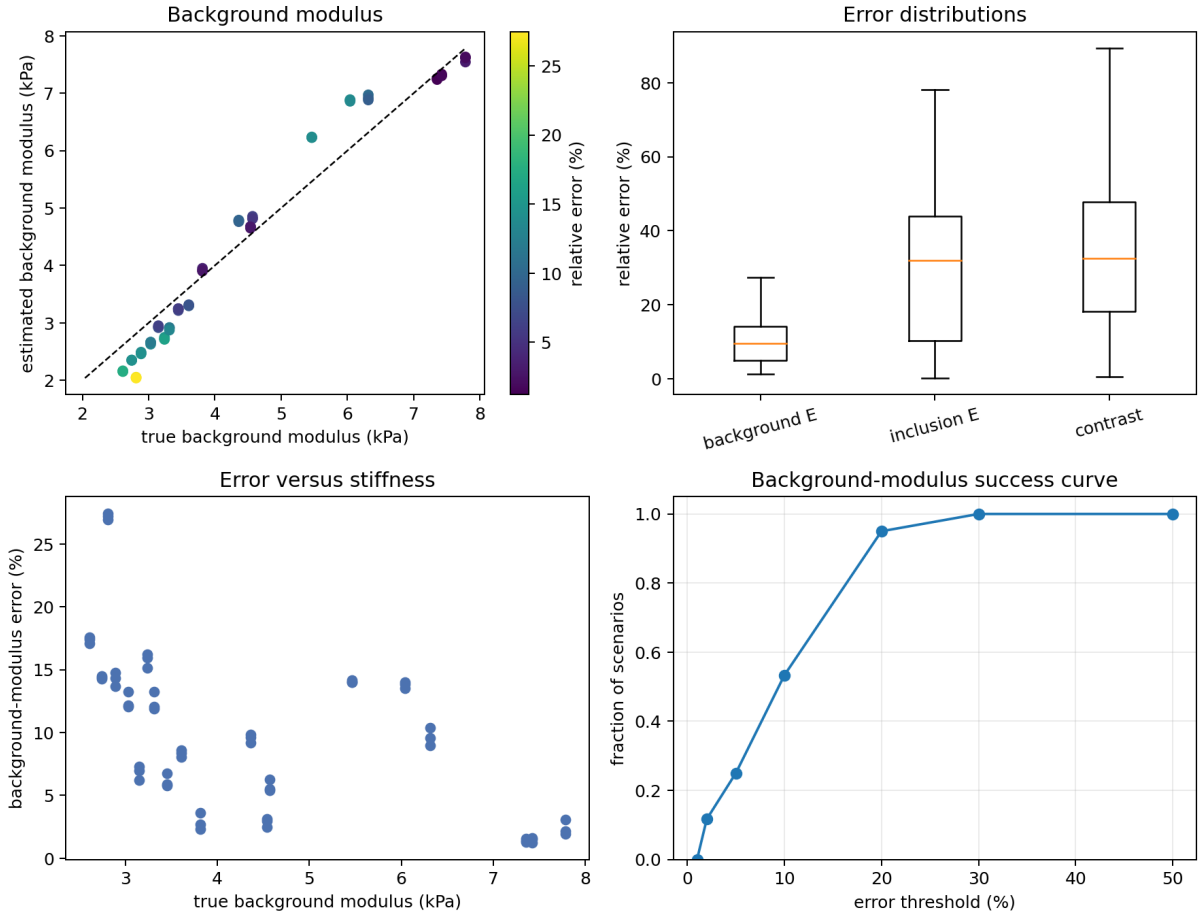


Fig. 4 Complete 60-scenario FEM inversion campaign. The panels report estimated versus true background modulus, error distributions for background modulus, inclusion modulus, and contrast, error versus true stiffness, and the cumulative background-modulus success curve.

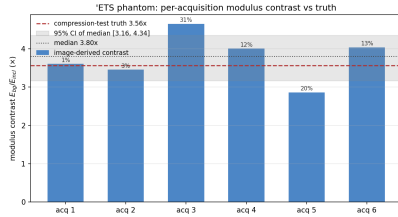


Fig. 5 ÉTS phantom: per-acquisition image-derived modulus contrast vs the compression-test truth (dashed), with the bootstrap 95% CI band and median. The video-derived contrast is consistent across all six acquisitions.

deformation and loading are available. The force-provenance analysis (Sec. 3.3) confirms that

uncertainty in the force scale propagates to absolute modulus but largely cancels in regional stiffness contrast. Absolute modulus and stiffness contrast should therefore be interpreted separately.

Example provenance record

For the pulling sequence, albedo and roughness are marked estimated because they are fitted to image observations. Tissue class is also marked estimated because it is selected by an optical heuristic. Young's modulus is marked assumed (liver prior: median 6 kPa; 95% interval 2.3–16.0 kPa) because no force was instrumented. Regional stiffness contrast is marked estimated from deformation, while Poisson's ratio and viscosity retain their table-derived assumed labels. Simulations based on this

Table 5 Mechanical inversion under successively weaker force evidence. The remaining rows are additional checks on the finite-element benchmark.

Setting	Evidence	Result
Synthetic closed loop (mass-spring)	uniform E	0.01% error
Synthetic inclusion (mass-spring)	4.0× contrast, surface-only	3.19×
Independent FEM benchmark	E_{bg} , measured force; noisy motion	median 9.6% error (60 scenarios)
Held-out load case (FEM)	press.left → other cases	within 4.1% of oracle
Noise robustness (FEM)	+0.5 / +1.0 mm noise	E_{bg} error 13–20% / 9–23%
Surrogate comparison	mass-spring versus FEM	82% error ⇒ FEM retained
Posterior width	Laplace approximation	inclusion flagged unidentified
Force provenance	auxiliary video-force error model → E	E_{bg} 4.1±4.0%; contrast unchanged
Real phantom (ÉTS)	contrast versus compression-test reference	6.9% error; 95% confidence interval includes reference

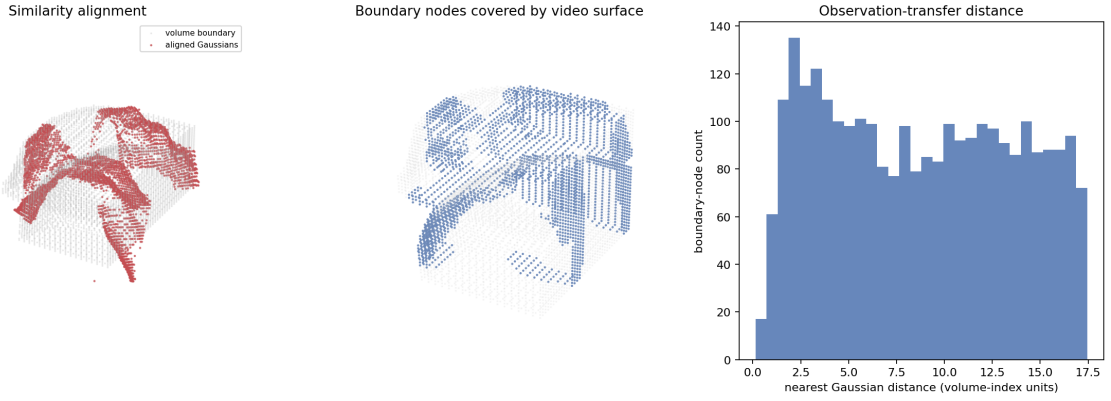


Fig. 6 Cross-subject computational mapping from the tracked Gaussian surface to the patient-specific CT-derived tetrahedral volume. Similarity alignment is followed by local interpolation onto covered boundary nodes. The mapping contains 28,523 nodes, 149,028 tetrahedra, and 2,755 mapped boundary nodes (38.7% coverage). Because the endoscopic and CT data are from different subjects and CT spacing is unavailable, this demonstrates the software interface rather than anatomical correspondence or clinical validity.

Table 6 Ablations.

Ablation	Variant	Metric
Mask supervision	instrument mask	11.13 dB
Mask supervision	depth-valid tissue mask	36.79 dB
Number of Gaussians	4000 / 6498 / 10000	35.31 / 36.86 / 38.21 dB
Temporal-model	inertia-only	37.23 (divergence)
Temporal-model	constant velocity + rollback	1.55
Parameterization	per-node field	contrast 1.69×
Parameterization	two-region	3.19×
Forward model	mass-spring surrogate	contrast 4.19× (true 1.71×)
Forward model	FEM-in-the-loop	2.26× (true 1.71×)

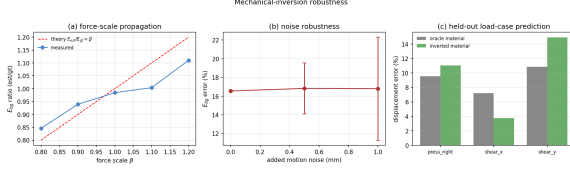


Fig. 7 Mechanical-inversion robustness. (a) Force-scale sweep: the recovered E_{bg} ratio tracks the theoretical one-to-one propagation of the force-scale relation. (b) Noise robustness: the E_{bg} error increases with added motion noise. (c) Held-out load-case prediction: the inverted material matches the oracle material across three unseen load cases.

sequence should therefore propagate the prior interval rather than use a single absolute modulus.

The proposed representation could support simulation of alternative instrument loads after reconstruction from an intra-operative video. However, the present evidence establishes technical feasibility rather than clinical utility. Force-instrumented tissue experiments, prospective patient data, and task-specific validation are required before the method can be considered for surgical decision support. Elastography provides the closest clinical analogue because it also estimates stiffness contrast from induced deformation [21]. The phantom experiment indicates agreement with independent compression testing for this relative quantity; it does not establish equivalence with clinical elastography.

Any future clinical implementation would also require institutional data governance, calibrated confidence thresholds, and regulatory evaluation. The provenance map provides a technical mechanism for exposing the evidence associated with each parameter, but does not by itself establish clinical safety.

Reproducibility

Source code, evaluation scripts, and data-split definitions are available in the project repository. Every numerical result reported in this paper is produced by the released end-to-end scripts, which cover reconstruction, tracking, and inversion on EndoNeRF and SCARED, the FEM benchmark inversion, the force-provenance analysis, and the phantom audit; each script writes a machine-readable summary and the corresponding figures, from which every number in the paper is drawn.

4.1 Limitations

The strict holdout shows that reconstruction accuracy is substantially below EndoGaussian; fixed surface identity and mechanical coupling do not compensate for the absence of a learned spatiotemporal deformation field. Although the FEM analysis covers all 60 predefined scenarios, the inclusion modulus and contrast remain weakly recovered. The auxiliary force estimator and its empirical error model are external to the core twin and have not been independently validated beyond their source recordings. The ÉTS phantom uses ultrasound rather than endoscopic colour images, so it validates mechanical contrast but not the optical reconstruction. The mass-spring surrogate has an 82% forward error and is not used to support absolute mechanical estimates on real data. In addition, the clinical CT experiment contains one de-identified case and validates geometry construction only. These limitations preclude claims of clinical performance or absolute-modulus recovery from uninstrumented endoscopic video.

5 Conclusion

The proposed twin couples a canonical Gaussian surface representation with fixed-correspondence tracking, force-aware mechanical parameterization, and parameter provenance. Tracked Gaussians provide co-registered surface motion and regional stiffness contrast for uninstrumented video; absolute modulus is recovered only in known-force FEM experiments. The force-scale identity explains why force uncertainty propagates to absolute modulus but cancels in regional contrast. The common holdout demonstrates that this mechanically coupled representation does not match the reconstruction accuracy of EndoGaussian. Across 60 FEM scenarios, median errors are 9.6% for background modulus, 32.0% for inclusion modulus, and 32.4% for contrast; the phantom contrast error is 6.9% against compression testing. The CT-derived tetrahedral mesh and explicit Gaussian-to-volume map demonstrate the software interface on one cross-subject case. These results support the observation-to-simulation framework as a computational method,

but do not establish clinical performance, state-of-the-art reconstruction, or force-free absolute-modulus recovery. Future work requires force-instrumented tissue experiments, paired patient-specific surface and volume data, and prospective evaluation of task-specific simulation accuracy.

Abbreviations

BRDF: bidirectional reflectance distribution function; CI: confidence interval; CT: computed tomography; FEM: finite-element method; MPM: material point method; NVS: novel-view synthesis; PSNR: peak signal-to-noise ratio; RGB: red, green, and blue; SCARED: Stereo Correspondence and Reconstruction of Endoscopic Data.

Declarations

Ethics approval and consent to participate

The public EndoNeRF, SCARED, finite-element benchmark, and ÉTS phantom datasets do not involve identifiable human subjects. The hospital computed tomography images are de-identified retrospective scans provided by Henan Provincial People’s Hospital for geometry-source validation; no identifiable patient information is reported. The name of the responsible ethics committee and the approval or waiver reference must be inserted after confirmation by the data-providing institution.

Consent for publication

Not applicable. The manuscript does not contain data from any identified individual person.

Availability of data and materials

Project name: Soft-tissue digital twin.
Project home page: <https://github.com/liranyang123456-commits/soft-tissue-digital-twin>.
Operating system(s): Platform independent.
Programming language: Python.
License: MIT for the released pipeline code.
The auxiliary R3D-18 force estimator used only to define the error model is external to the core framework and is not part of this contribution.

The code, evaluation scripts, data-split definitions, aggregate force-error parameters, and a README that generate the reported figures and tables are available in that repository. EndoNeRF, SCARED, the finite-element benchmark, and the ÉTS phantom are public datasets cited in this article. The per-sample predictions from the auxiliary force-estimation study are not part of this article; the aggregate parameters required for the propagation experiment are included in the repository. The de-identified hospital computed tomography images are not publicly available under the data-providing institution’s governance, but they are available from the corresponding author on reasonable request.

Competing interests

The authors declare that they have no competing interests.

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Authors’ contributions

RL conceived the study, developed the method, wrote the software, and drafted the manuscript. HM contributed the algorithm, software, and validation. ZL contributed validation, formal analysis, and multi-view evaluation. NW contributed clinical data curation and anatomical review. All authors read and approved the final manuscript.

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